

# Apply controlled mechanical strain during the thermal annealing of the borosilicate glass liner to induce grain boundary realignment and reduce sliding propensity.

## Executive Technology Brief

**Domain:** mechanical\_structural

**Technology Maturity:** ENGINEERING\_DEFINITION

**Transfer Posture:** Sponsored Validation

**Problem:** Documented problem 'grain-boundary sliding' concerning borosilicate glass reactor liner. User-submitted engineering problem (session input): "Why do grain-boundary sliding failures limit the service temperature of borosilicate glass reactor liners, and how can the grain structure be refined to fix this?". [cod\_optimade] retrieval status EMPTY — provider outcome recorded, not treated as absence (Art. XXI.3). [arxiv] 5 records retrieved live (2010-2025). Illustrative records: 'Motion of grain boundaries incorporating dislocation structure'; 'Energy of low angle grain boundaries based on continuum dislocation structure'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

**Mechanism:** Integrated advanced strain engineering (as used in CMOS transistors to modify crystal behavior)

**Architecture:** PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: Apply controlled mechanical strain during the thermal annealing of the borosilicate glass liner to induce grain boundary realignment and reduce sliding propensity.

**Governing Model:** Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

## Key Equations:

MEC-002: Section bending stress. Expression:  $\sigma = M * c / I$ . Applicability to this invention: CONDITIONAL; holds under: linear-elastic material; small deflection; prismatic section. Assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design. Engaged variables: NOT ESTABLISHED.

MEC-003: Hooke's law (linear elasticity). Expression:  $\sigma = E * \epsilon$ . Applicability to this invention: APPLICABLE; holds under: below proportional limit; isothermal. Assumptions: superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead. Engaged variables: epsilon.

MEC-001: Euler buckling load. Expression:  $P_{cr} = \pi^2 * E * I / (K * L)^2$ . Applicability to this invention: CONDITIONAL; holds under: elastic, slender column, small deflection. Assumptions: K for catheter constraint conditions is MODEL\_DERIVED. Engaged variables: NOT ESTABLISHED.

**Kill Condition:** killer experiment fails: falsification\_test\_from\_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING\_DEFINITION. No physical prototype. No clinical validation.

*Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.*